Donald Pinckney

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Education

Northeastern University, PhD in Computer Science Sep 2020 – Nov 2024
Focused on Dev Tools & AI Agents, GPA: 4.00, Advised by Drs. Arjun Guha and Jonathan Bell

University of Massachusetts Amherst, MS in Computer Science Sep 2018 – May 2020
Focused on Compilers & Cloud Infrastructure, GPA: 3.87, Advised by Drs. Arjun Guha and Yuriy Brun

University of California Davis, BS in Computer Science and Mathematics Sep 2014 – Jun 2018
Double Major in Computer Science & Engineering and Mathematics, GPA: 3.94

Skills

Languages: Rust, Python, C/C++, JavaScript/TypeScript, SQL, Haskell, OCaml.
Infrastructure & Systems: Google Cloud (Vertex AI, GCE, VPC), AWS (S3, ECS, AgentCore), Kubernetes, Docker, Temporal, gRPC, PostgreSQL, SQLite, Redis, E2B, Cloudflare Workers, CI/CD.
Agentic AI Ecosystem: LangChain/LangGraph, Google ADK (Agent Development Kit), OpenAI Agents SDK, Pydantic AI, AG-UI/CopilotKit, MCP (Model Context Protocol), Agent Evaluation Frameworks.
Specialties: Distributed Systems Architecture, Agentic Infrastructure, Program Analysis/Static Analysis, Formal Verification, API/SDK Design.

Experience

Sectr (sectr.dev), *Founder & CEO* Aug 2026 – Present

- Founded Sectr, the data and coordination layer for AI agents: it stores, streams and resumes agent conversations, so teams can turn agents into products with no rewrite and no framework lock-in. Went from problem discovery (20+ interviews) to a free public beta in about two months, solo.
- Architected the Rust control plane and data plane (100k+ lines), with a durable journal as the single source of truth (Postgres in the cloud, SQLite locally), so agents suspend, scale to zero, and resume exactly from the transcript, seconds or weeks later, e.g. while a person approves a tool call.
- Built framework adapters for the OpenAI Agents SDK, LangGraph and Pydantic AI (a two-line integration), and a per-agent AG-UI endpoint that streams tokens to CopilotKit frontends.
- Shipped sandboxed execution on E2B with injected Rust sidecars, a CLI and local dev server, and a Next.js dashboard and docs on Cloudflare Workers.

Temporal Technologies, *Senior Software Engineer (AI Foundations)* Sep 2025 – Aug 2026

- Spearheaded the technical design of AI-native SDK primitives in Python, collaborating with cross-functional stakeholders to meet customers' evolving needs with the Temporal stack in the domain of AI.
- Led prototyping of integrations between Temporal and AWS AgentCore runtime, extending the functional lifespan of session-isolated runtimes to support long-running asynchronous agent tasks. Collaborated with peers and AWS engineers to write and review educational and messaging content.
- Participated in design discussions for integrations between Temporal and the Google Agent Development Kit (ADK), enabling cloud-based agents to be both durable and resumable for long-running tasks.
- Developed a flexible plugin system for Temporal SDKs to meet the needs of peers and stakeholders to significantly increase internal developer velocity when creating integrations with 3rd party agent SDKs.

Gitar (venture-backed startup), *Founding AI Coding Agent Engineer* Apr 2024 – Sep 2025

- Launched Gitar, the first coding agent leveraging deep static analysis, resulting in a 700% spike in customer demand and successful enterprise pilots.
- Built agent tools for automated program transformations (symbol lookup, constant folding, dead code analysis), enabling precise and safe code changes at scale.
- Co-designed cloud infrastructure to support remote execution of Gitar, prioritizing safety via customer data isolation and a fine-grained permission system to validate external API calls by the agent.
- Developed a synthetic data generation framework for evaluating agentic reasoning and testing correctness of refactoring tools.

Stealth Startup (Reactive PaaS), *Founder* Jan 2023 – Apr 2024

- Designed a novel application programming model for distributed and event-driven systems by unifying state management with function orchestration, allowing developers to define reactive microservices that automatically react to synchronized state.
- Engineered a high-performance content-addressable state versioning engine that computes and distributes diffs among microservice actors, minimizing network overhead by eliminating redundant data transfers.

Northeastern University, Programming Research Lab, *Dev Tools & AI Agents PhD* Sep 2020 – Nov 2024

- Built agentic package managers that integrate symbolic optimization reasoning with LLM reasoning to automatically apply dependency patches based on runtime errors, avoid packages with security vulnerabilities, minimize code size, and more.
- Architected a large distributed system to scrape and archive the global NPM package registry (36M+ packages, 20+ TB) in real-time, utilizing an append-only, N-sharded architecture to enable lock-free concurrent writes and unlimited reads across hundreds of HPC nodes while optimizing for NFS caching and consistency models. Mentored an undergraduate on the implementation and led technical design reviews to resolve performance bottlenecks as ingestion load increased.
- Developed a novel methodology (MultiPL-E) to automatically translate the HumanEval and MBPP benchmarks, enabling multilingual evaluation of LLMs. MultiPL-E now supports 31 different evaluation languages and is used actively on the BigCode Models Leaderboard on HuggingFace.

Uber, Programming Systems Group, *Dev Tools Research Scientist* May 2020 – Dec 2020

University of Massachusetts Amherst, *Compilers & Cloud Infrastructure MS* Sep 2018 – May 2020

- Pioneered the study of formal semantics for serverless computing (FaaS), laying a theoretical foundation for cloud providers to develop new FaaS abstractions, such as Microsoft Azure's Durable Functions.
- Reduced code size by 23% and sped up programs by 15% for multithreaded WebAssembly by extending a JIT compiler (Wasmtime) with stack capture and resume instructions at the virtual machine level.

Apple Inc., *macOS Frameworks Team, Intern* Summers 2014, 2015, 2016

Publications

PhD Dissertation 2024 *Improving Dependency Management via Formal Semantics.* **Donald Pinckney**.
[PDF]. Keywords: LLMs, Agents, Formal Methods, Optimization, Empirical Methods, Evaluations.

ICSE 2023 (Acceptance rate: 15%) *Flexible and Optimal Dependency Management via Max-SMT.* **Donald Pinckney**, Federico Cassano, Arjun Guha, Jonathan Bell, Massimiliano Culpo, Todd Gamblin.
[paper] [talk] [github] [install]. Keywords: Program Analysis, Empirical Methods, Theoretical Frameworks.

MSR 2023 *A Large Scale Analysis of Semantic Versioning in NPM.* **Donald Pinckney**, Federico Cassano, Arjun Guha, Jonathan Bell.
[paper]. Keywords: Empirical Methods, Data Collection, Data Analysis.

ESEC/FSE 2023 Demo Track *npm-follower: A Complete Dataset Tracking the NPM Ecosystem.* **Donald Pinckney**, Federico Cassano, Arjun Guha, Jonathan Bell.
[paper] [talk] [dataset] [github]. Keywords: Data Collection, Data Analysis.

TSE 2023 *MultiPL-E: A Scalable and Polyglot Approach to Benchmarking Neural Code Generation.* **Donald Pinckney**, Federico Cassano, John Gouwar, Daniel Nguyen, Sydney Nguyen, Luna Phipps-Costin, Ming-Ho Yee, Yangtian Zi, Carolyn Jane Anderson, Molly Q Feldman, Arjun Guha, Michael Greenberg, Abhinav Jangda.
[paper] [talk] [github] [website]. Keywords: LLMs, Empirical Methods, Model Benchmarking, Data Generation.

DLS 2020 *Wasm/k: Delimited Continuations for WebAssembly.* **Donald Pinckney**, Yuriy Brun, Arjun Guha.
[paper] [talk] [github] [website]. Keywords: Compilers, Program Analysis, Empirical Methods, Performance Benchmarking.

OOPSLA 2019 (Acceptance rate: 15%), Distinguished Paper Award *Formal Foundations of Serverless Computing.* Abhinav Jangda, **Donald Pinckney**, Yuriy Brun, Arjun Guha.
[paper] [talk] [website]. Keywords: Cloud Infrastructure, Empirical Methods, Theoretical Frameworks, Performance Benchmarking.